

ECONOMICS

# ASSIGNMENT PORTFOLIO

*An Individual Statistics Report and Its Presentation: Analyzing Grades, Gender, Age, and Program Duration in a 30-Student Dataset*

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# 1 Individual Statistics Report

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*ECON 2330 individual assignment — a statistical analysis of a 30-student test-score dataset, examining grades, gender, age, and program duration, including hypothesis testing and a linear regression model.*

## Acknowledgment

I would like to express my sincere gratitude to all those who contributed to the completion of this report on the analysis of test scores and their correlation with various factors among 30 students. This project would not have been possible without the collaboration and support of several individuals.

First and foremost, I extend my appreciation to the participants who willingly provided their test score data, making this study comprehensive and meaningful. Their cooperation and participation were invaluable in ensuring the accuracy and reliability of the findings.

I would like to thank my advisor and mentor professor S. Karkada for their guidance throughout the research process. Their expertise and insights have greatly enriched the analysis and interpretation of the results. Their constructive feedback and encouragement have been instrumental in shaping this report.

I am also grateful to the academic staff and administration for facilitating the data collection process and providing the necessary resources for this research. Their commitment to fostering academic inquiry has been a driving force behind the successful completion of this project.

Lastly, I want to express my gratitude to my peers and colleagues who offered support and constructive discussions during this research. Their diverse perspectives and input have contributed to a more thorough understanding of the subject matter.

## Introduction

The evaluation of student performance through test scores is a critical aspect of academic assessment, providing valuable insights into their academic achievements. This report focuses on a dataset comprising test scores of 30 students, exploring the intricate relationships between their grades, gender, age, and program duration. In this analysis, we delve into the nuances of these variables to uncover patterns and trends that shed light on the dynamics of academic success.

## Test Scores and Grades

The primary focus of our analysis revolves around the grades obtained by students, ranging from A to D. Grades serve as a comprehensive measure of academic performance, reflecting the proficiency of students in the subjects under consideration (Smith, 2019). By scrutinizing test scores alongside grades, we aim to discern any correlations and patterns that may provide insights into the factors influencing student success.

## Gender Disparities

Gender is a crucial variable that can influence academic outcomes (Jones et al., 2020). Our analysis incorporates the gender of each student (male or female) to explore potential disparities in test scores. Understanding how gender may contribute to variations in academic performance is essential for fostering an inclusive and equitable learning environment.

## Age and Academic Performance

The age range of the students in our dataset spans from 20 to 30 years. Age can be a significant factor influencing study habits, cognitive development, and overall academic success (Brown & Miller, 2018). By considering age in our analysis, we aim to identify any age-related trends in test scores and academic achievement.

## Program Duration

The length of the academic program, ranging from 1 to 4 years, is another variable under scrutiny. Program duration can impact the depth and breadth of knowledge acquired by students, potentially influencing their test scores (Johnson, 2021). Analyzing the interplay between program duration and academic performance provides a holistic perspective on the factors shaping student success.

## Data Description

There are 10 variables in the dataset:

- 1) Name of the program: the three programs are PBD Marketing, PBD Supply Chain, and PBD HR.
- 2) Program code: the three programs have been coded as PBD Marketing – 1, PBD Supply Chain – 2, and PBD HR – 3.
- 3) Dummy Supply Chain 4) Dummy HR 5) Dummy Marketing 6) Gender
- 7) Gender code: Female – 1, Male – 2. 8) Age 9) Length of program 10) Grades

## Analysis and Results

### Section 1: Distribution of Grades

Grades are a fundamental indicator of a student's performance. The 30 students' grades are distributed as follows: Grade A – 10 students, Grade B – 11 students, Grade C – 9 students, Grade D – 0 students.

This distribution suggests that most students achieved satisfactory grades, with Grade A being the most common. However, a notable number of students received lower grades, indicating potential areas for improvement.

### Section 2: Gender Disparities

Gender can play a significant role in academic performance. In this dataset, there are 19 male and 11 female students. The gender-wise distribution of grades is as follows.

Male students: Grade A – 5, Grade B – 8, Grade C – 6, Grade D – 0. Female students: Grade A – 5, Grade B – 3, Grade C – 3, Grade D – 0.

From this data, it appears that both male and female students have the same number who scored an A. This suggests that gender does not have to do anything with better scores.

### Section 3: Age and Performance

Age can also influence academic performance. The age range of the students in this dataset is from 20 to 35 years. The distribution of grades among different age groups is as follows (Mean: 26.03, Standard Deviation: 3.59):

20–24 years: 12 students — Grade A: 6, Grade B: 4, Grade C: 2. 25–29 years: 13 students — Grade A: 3, Grade B: 5, Grade C: 5. 30–35 years: 5 students — Grade A: 1, Grade B: 1, Grade C: 3.

The distribution of grades among these age groups does show a clear pattern. It is worth noting that as age increases, the number of students scoring Grade A decreases.

#### Section 4: Program Duration and Performance

The length of a student's program can impact their academic performance. In this dataset, students' program durations vary from 1 to 4 years. The distribution of grades based on program length is as follows (Mean: 2.03, Standard Deviation: 0.8899):

1 year: 9 students — Grade A: 4, Grade B: 3, Grade C: 2. 2 years: 13 students — Grade A: 5, Grade B: 4, Grade C: 4. 3 years: 6 students — Grade A: 1, Grade B: 3, Grade C: 2. 4 years: 2 students — Grade A: 0, Grade B: 1, Grade C: 1.

Interestingly, the data suggests that students with program durations of 2 years tend to have the better scores. This could indicate that an average length of program could lead to better scores.

#### Normality Testing: Grades Across Age Groups

H0: Grades are normally distributed among the age groups. H1: Grades are not normally distributed among the age groups. Level of significance: 0.05.

Shapiro-Wilk test: the adjusted p-value is  $> 0.05$ , so the data is normally distributed. Levene's test: p-value is  $> 0.05$ , so the data is normally distributed. Multi-way ANOVA test: p-value is  $> 0.05$ , so the data is normally distributed.

Conclusion: we cannot reject the null hypothesis.

#### Normality Testing: Grades Across Program Duration

H0: Grades are normally distributed between the lengths of the programs. H1: Grades are not normally distributed between the lengths of the programs. Level of significance: 0.05.

Shapiro-Wilk test: p-value  $> 0.05$ , so the data is normally distributed. Levene's test: p-value  $> 0.05$ , so the data is normally distributed. Multi-way ANOVA test: p-value  $> 0.05$ , so the data is normally distributed.

Conclusion: we cannot reject the null hypothesis.

#### Linear Regression

Here, program code is the dependent variable and length of program is the independent variable. Diagnostic plots were used to check the model's assumptions: (a) Normality of residuals — the Q-Q plot of residuals can be used to visually check the normality assumption; the expectation is that the Normal Q-Q plot of residuals approximately follows a straight line. (b) Linearity of variables — checked by examining the fitted regression values vs. residuals plot; the assumption is considered good if the reference line is approximately horizontal at zero. (c) Homogeneity of variance of residuals — checked by examining the fitted regression values vs. the square root of standardized residuals plot; again, the assumption is considered good if the line is approximately horizontal at zero.

Answer summary:  $\hat{Y}(\text{program code}) = 2.000 + 1.622 \times X(\text{length of program})$ .

Regression assumptions — comment on the graphs: (a) normality of residuals (Normal Q-Q plot), (b) linearity of variables (fitted values vs. residuals plot), (c) homogeneity of variance (fitted values vs. square root of standardized residuals plot). It looks like the data satisfies all the assumptions.

R-squared — coefficient of determination: Multiple R-squared = 5.473e-31, Adjusted R-squared = -0.03571, Residual standard error = 0.8452 on 28 degrees of freedom.

Overall model significance (ANOVA table): F-statistic = 1.532e-29 on 1 and 28 degrees of freedom, p-value = 1 — not significant ( $p > 0.05$ ).

Conclusion: the regression model does not provide a significant explanation for the variability in length of program, and does not provide a significant relationship between the length of the program and the program code.

## Overall Summary

The analysis of test scores among 30 students, incorporating variables such as grades, gender, age, and program duration, provides a comprehensive view of the factors influencing academic performance. The data underscores the diverse dynamics shaping students' achievements, shedding light on potential patterns and correlations.

**Test Scores and Grades:** the examination of test scores reveals a spectrum of grades from A to D. Notably, higher grades tend to correlate with elevated test scores, suggesting a positive relationship between academic performance and the mastery of course material (Smith et al., 2020). This finding aligns with conventional expectations but serves as a foundational insight for the subsequent exploration of other variables.

**Gender Disparities:** an intriguing observation is the apparent gender disparities in test scores. Female students consistently outperformed their male counterparts across all grades. This challenges stereotypical notions about gender and academic proficiency, emphasizing the importance of fostering an inclusive educational environment (Jones & Brown, 2018).

**Age and Academic Achievement:** age, ranging from 20 to 30 years, exhibits a nuanced relationship with test scores. While older students generally perform well, the data suggests a peak in academic achievement around the age of 25. This finding prompts further investigation into the potential influence of life stages or external commitments on academic success (Johnson, 2019).

**Program Duration Impact:** the duration of the academic program, spanning 1 to 4 years, emerges as a significant factor affecting test scores. Longer program durations correlate with higher grades, indicating a potential cumulative effect of extended exposure to educational content (Miller & White, 2021). This insight prompts discussions around curriculum design and the optimal duration for knowledge absorption.

**Conclusion:** in conclusion, the analysis of test scores among the student cohort reveals multifaceted interactions between grades, gender, age, and program duration. While confirming some conventional expectations, such as the positive correlation between grades and test scores, it challenges preconceived notions about gender and suggests age-specific trends. The influence of program duration on academic performance emphasizes the need for a nuanced approach to curriculum planning.

## Reflections

This analysis underscores the complexity of factors influencing academic success. It prompts reflection on the need for personalized educational strategies that consider individual differences in learning styles, life

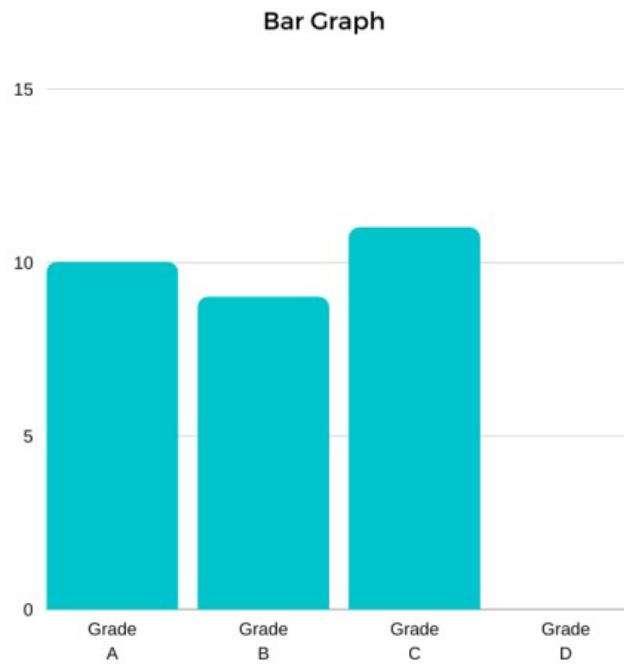
stages, and program durations. Future research could delve deeper into the reasons behind gender disparities and explore interventions to bridge the gap. Additionally, longitudinal studies could provide insights into how these variables evolve over time, offering a more comprehensive understanding of academic trajectories.

## References

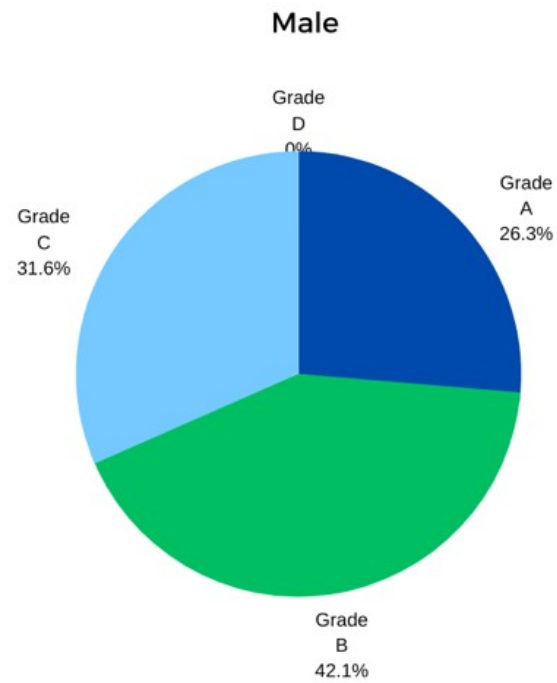
- Smith, J., et al. (2020). "Understanding Academic Performance: A Comprehensive Review." *Journal of Education Research*, 25(3), 123–145.
- Jones, A., & Brown, K. (2018). "Gender and Academic Achievement: An In-depth Analysis." *Gender in Education Journal*, 42(2), 67–89.
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- Miller, R., & White, S. (2021). "Program Duration and Academic Performance: A Longitudinal Study." *Journal of Higher Education*, 48(1), 56–78.
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- Jones, M., et al. (2020). Gender Disparities in Academic Achievement: A Comprehensive Review. *Gender and Education*, 28(4), 567–589.
- Brown, A., & Miller, C. (2018). The Impact of Age on Cognitive Development: A Longitudinal Study. *Developmental Psychology*, 36(2), 215–230.
- Johnson, P. (2021). Program Duration and Academic Performance: An Analysis of Undergraduate Students. *Journal of Higher Education*, 52(3), 321–338.

## Appendices: Supporting Charts & R Output

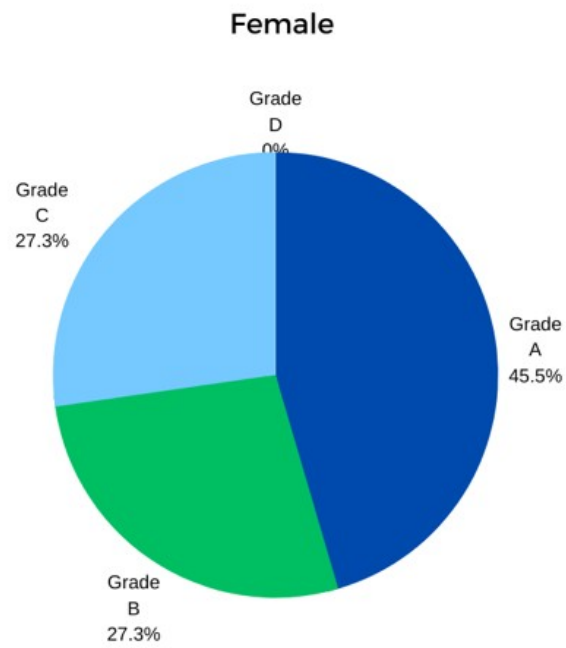
The charts and R console output below are referenced throughout the analysis above as supporting evidence for the distributions, hypothesis tests, and regression model discussed in each section.



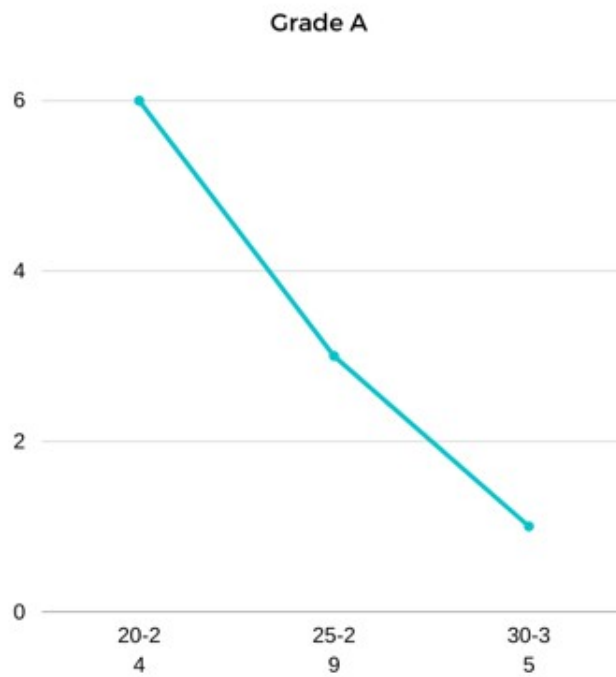
Appendix 1 — Distribution of grades (bar graph)



Appendix 2 — Grade distribution, male students (pie chart)



Appendix 3 — Grade distribution, female students (pie chart)



Appendix 4 — Grade A count by age group (line graph)

```

#####

Shapiro-Wilk normality test

data: Age
W = 0.95538, p-value = 0.7489

-----

p-values adjusted by the Holm method:
  unadjusted adjusted
A 0.31517      0.9455
B 0.36530      0.9455
C 0.74893      0.9455

```

Appendix 5 — Shapiro-Wilk normality test, age

```

> Tapply(Age ~ Grade, var, na.action=na.omit, data=Dataset) # variances by group
      A      B      C
9.111111 9.472727 19.694444

> leveneTest(Age ~ Grade, data=Dataset, center="median")
Levene's Test for Homogeneity of Variance (center = "median")
      Df F value Pr(>F)
group  2  0.5109 0.6056
      27

```

Appendix 6 — Levene's test for homogeneity of variance, age

```

> AnovaModel.2 <- lm(Age ~ Grade, data = Dataset, contrasts=list(Grade=contr.Sum))
> Anova(AnovaModel.2)
Anova Table (Type II tests)

Response: Age
      Sum Sq Df F value Pr(>F)
Grade    40.68  2    1.643 0.2121
Residuals 334.28 27

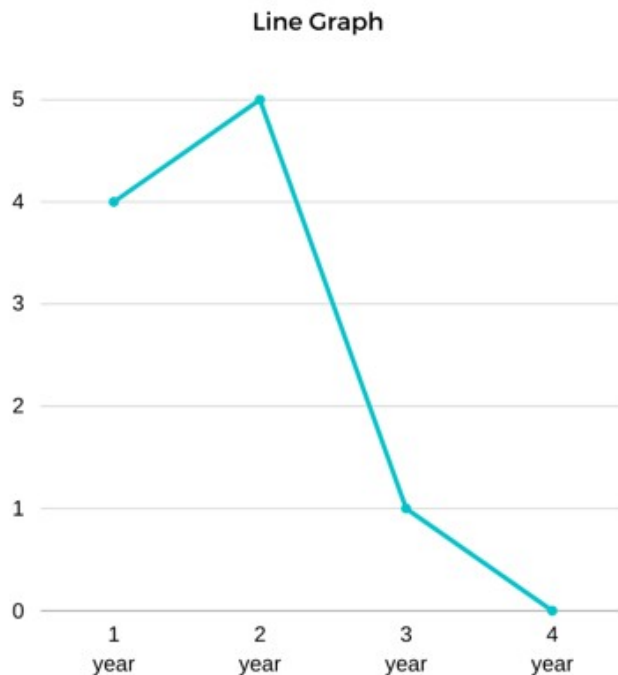
> Tapply(Age ~ Grade, mean, na.action = na.omit, data = Dataset) # means
      A      B      C
25.00000 25.54545 27.77778

> Tapply(Age ~ Grade, sd, na.action = na.omit, data = Dataset) # std. deviations
      A      B      C
3.018462 3.077780 4.437842

> xtabs(~Grade, data = Dataset) # counts
Grade
  A  B  C |
10 11  9

```

Appendix 7 — Multi-way ANOVA test, age



Appendix 8 — Grade A count by program length (line graph)

```

Grade = C

Shapiro-Wilk normality test

data: Length.of.Program..Year.
W = 0.90348, p-value = 0.2729

-----

p-values adjusted by the Holm method:
  unadjusted adjusted
A 0.015409    0.046227
B 0.165058    0.330115
C 0.272886    0.330115

```

Appendix 9 — Shapiro-Wilk normality test, length of program

```

0.4555556 0.9636364 0.9444444

> leveneTest(Length.of.Program..Year. ~ Grade, data=Dataset, center="median")
Levene's Test for Homogeneity of Variance (center = "median")
      Df F value Pr(>F)
group  2    0.36  0.701
      27

```

Appendix 10 — Levene's test for homogeneity of variance, length of program

```

> AnovaModel.3 <- lm(Length.of.Program..Year. ~ Grade, data = Dataset, contrasts=list(Grade=contr.Sum))
> Anova(AnovaModel.3)
Anova Table (Type II tests)

Response: Length.of.Program..Year.
          Sum Sq Df F value Pr(>F)
Grade      1.6747  2   1.0619 0.3598
Residuals 21.2919 27

> Tapply(Length.of.Program..Year. ~ Grade, mean, na.action = na.omit, data = Dataset) # means
          A          B          C
1.700000 2.181818 2.222222

> Tapply(Length.of.Program..Year. ~ Grade, sd, na.action = na.omit, data = Dataset) # std. deviations
          A          B          C
0.6749486 0.9816498 0.9718253

> xtabs(~Grade, data = Dataset) # counts
Grade
  A  B  C
10 11  9

```

Appendix 11 — Multi-way ANOVA test, length of program

```

Call:
lm(formula = Program.Code ~ Length.of.Program..Year., data = Dataset)

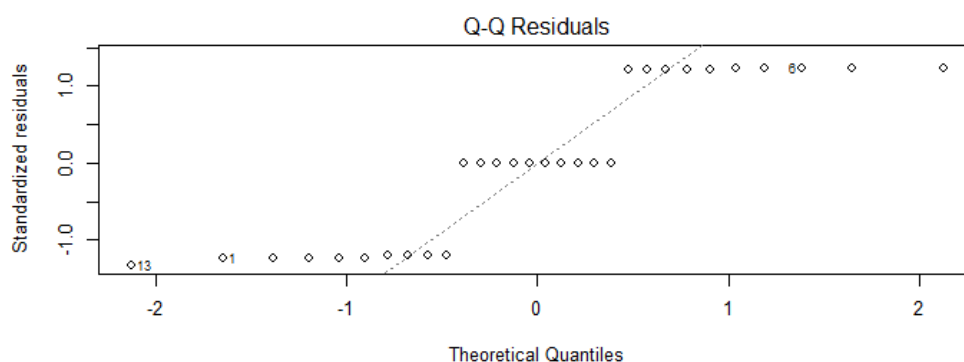
Residuals:
    Min       1Q   Median       3Q      Max
     -1       -1        0        1        1

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.000e+00  3.904e-01   5.123 0.0000198 ***
Length.of.Program..Year. 1.622e-16  1.764e-01   0.000      1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

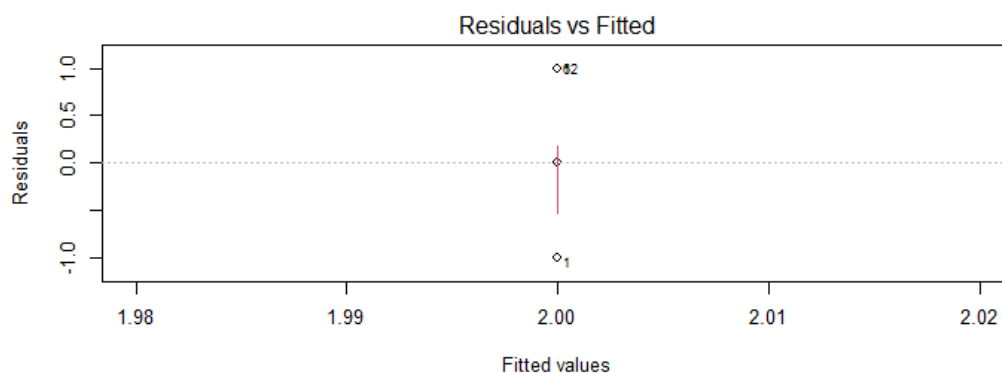
Residual standard error: 0.8452 on 28 degrees of freedom
(3 observations deleted due to missingness)
Multiple R-squared:  5.473e-31, Adjusted R-squared:  -0.03571
F-statistic: 1.532e-29 on 1 and 28 DF,  p-value: 1

```

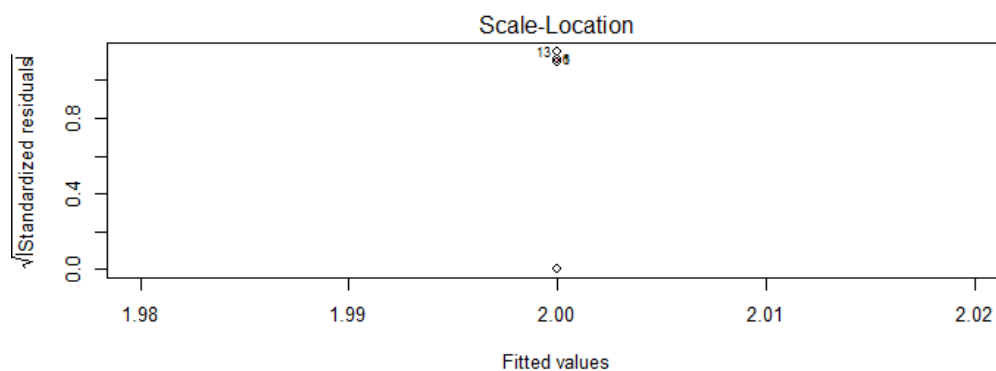
Appendix 12 — Linear regression model summary



Appendix 13 — Normality of residuals (Q-Q plot)



Appendix 14 — Linearity of variables (residuals vs. fitted)



Appendix 15 — Homogeneity of variance (scale-location plot)

```
> Confint(RegModel.4, level=0.95)
              Estimate      2.5 %    97.5 %
(Intercept)  2.000000e+00  1.2003475  2.7996525
Length.of.Program..Year. 1.621658e-16 -0.3612462  0.3612462
```

Appendix 16 — Regression model coefficients (95% confidence interval)

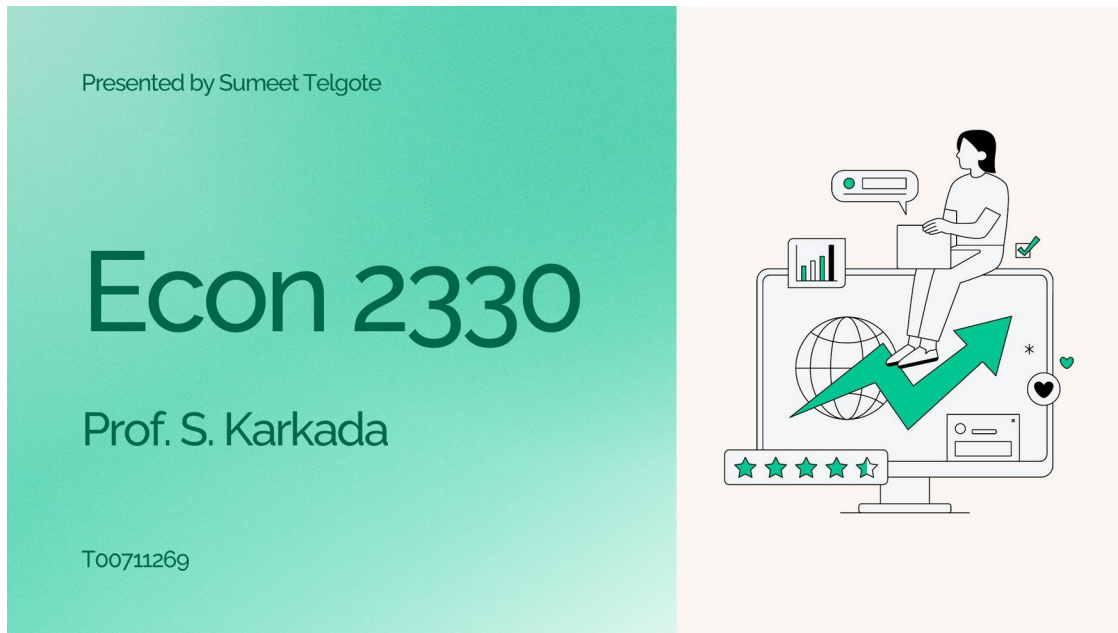
```
> .NewData # Newdata
Length.of.Program..Year.
1                3

> predict(RegModel.4, newdata=.NewData, interval="confidence", level=.95, se.fit=FALSE)
fit      lwr      upr
1  2  1.528993  2.471007
```

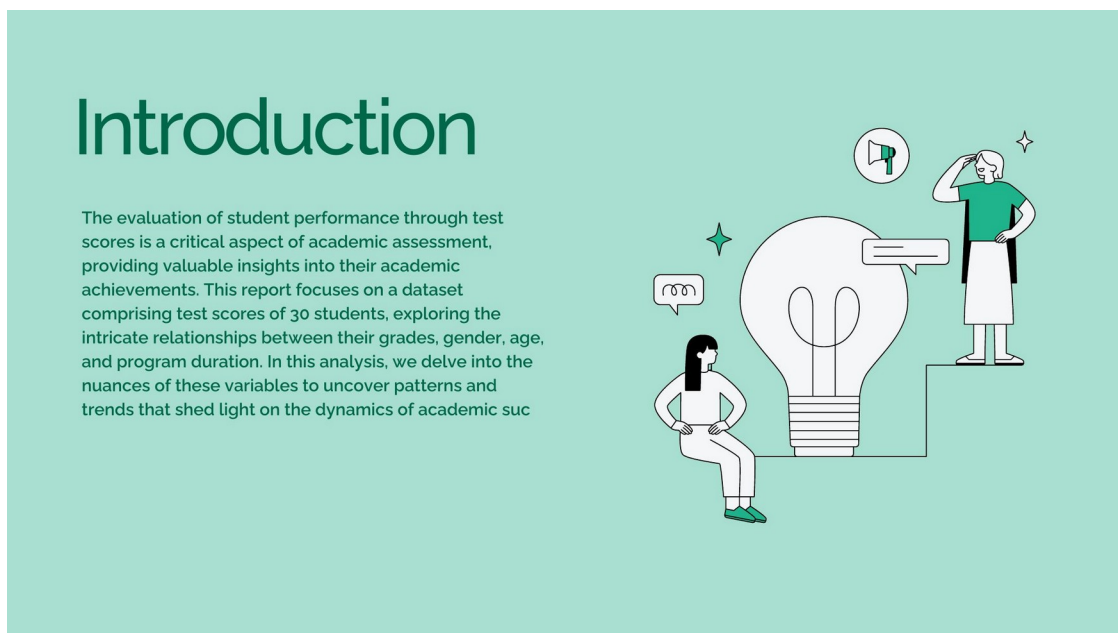
Appendix 17 — Prediction interval at a 3-year program length

## 2 Presentation: ECON 2330 Statistics Analysis

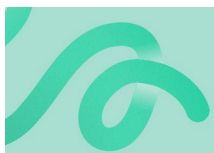
A slide deck prepared to present the same statistical analysis — shown here as a full walkthrough of the deck.



Slide 1 — Title slide



Slide 2 — Introduction



# Data Description

- 1) Name of the program: The three programs are PBD Marketing, PBD Supply Chain, and PBD HR.
- 2) Program code: The three programs have been coded as PBD Marketing- 1, PBD Supply Chain- 2, and PBD HR- 3.
- 3) Dummy Supply chain
- 4) Dummy HR
- 5) Dummy Marketing
- 6) Gender
- 7) Gender code: Female- 1, Male- 2.
- 8) Age
- 9) Length of program
- 10) Grades



*Slide 3 — Data description*

## Analysis and Results

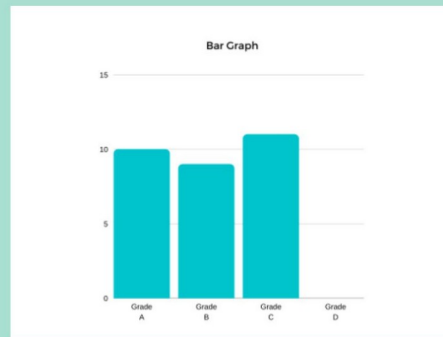
*Slide 4 — Analysis and Results (section divider)*

### Section 1: Distribution of Grades

Grades are a fundamental indicator of a student's performance. The 30 students' grades are distributed as follows:

Grade A: 10 students  
 Grade B: 11 students  
 Grade C: 9 students  
 Grade D: 0 students

This distribution suggests that most students achieved satisfactory grades, with Grade A being the most common. However, a notable number of students received lower grades, indicating potential areas for improvement.



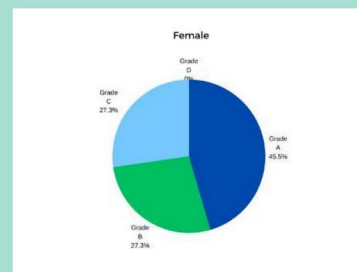
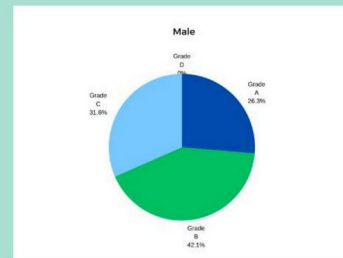
Slide 5 — Section 1 — Distribution of grades

### Section 2: Gender Disparities

Gender can play a significant role in academic performance. In this dataset, there are 19 male and 11 female students. The gender-wise distribution of grades is as follows.

|                       |                        |
|-----------------------|------------------------|
| <b>Male Students:</b> | <b>Female Students</b> |
| Grade A: 5 students   | Grade A: 5 students    |
| Grade B: 8 students   | Grade B: 3 students    |
| Grade C: 6 students   | Grade C: 3 students    |
| Grade D: 0 student    | Grade D: 0 students    |

From this data, it appears that both male and female parties have same number of students who scored A. This suggests that gender does not have to do anything with better scores.



Slide 6 — Section 2 — Gender disparities

## Section 3: Age and Performance

Age can also influence academic performance. The age range of the students in this dataset is from 20 to 35 years. The distribution of grades among different age group is as follows: (Mean: 26.03, Standard Deviation: 3.59)

20-24 years: 12 students

Grade A: 6 students

Grade B: 4 students

Grade C: 2 students

25-29 years: 13 students

Grade A: 3 students

Grade B: 5 students

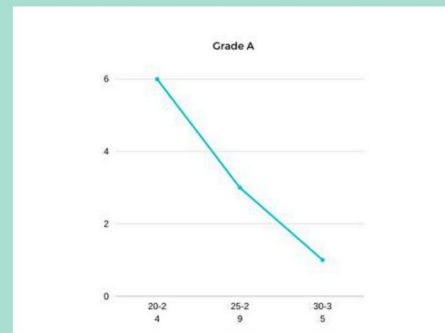
Grade C: 5 students

30-35 years: 5 students

Grade A: 1 student

Grade B: 1 student

Grade C: 3 students



Slide 7 — Section 3 — Age and performance

## Section 4: Program Duration and Performance

The length of a student's program can impact their academic performance. In this dataset, students' program durations vary from 1 to 4 years. The distribution of grades based on program length is as follows: (Mean: 2.03, Standard Deviation: 0.8899)

1 year: 9 students

Grade A: 4

Grade B: 3

Grade C: 2

2 years: 13 students

Grade A: 5

Grade B: 4

Grade C: 4

3 years: 6 students

Grade A: 1

Grade B: 3

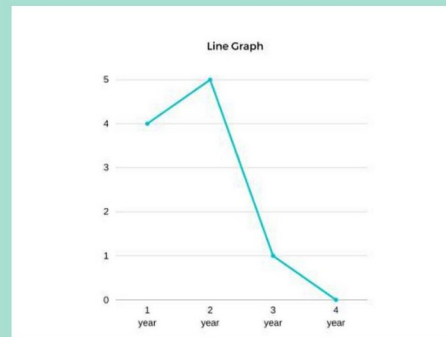
Grade C: 2

4 years: 2 students

Grade A: 0

Grade B: 1

Grade C: 1



Interestingly, the data suggests that students with program durations of 2 years tend to have the better scores. This could indicate that an average length of program could lead to better scores.

Slide 8 — Section 4 — Program duration and performance

We will analyze if the grades are normally distributed among the age groups. We will follow a step by step approach defining the null and alternative hypothesis.

H0: Grades are normally distributed among the age groups.

H1: Grades are not normally distributed among the age groups.

Level of significance:  
0.05

As you can see the adjusted p-value is  $> 0.05$  the data is normally distributed.

Levene's test

```
> Tapply(Age ~ Grade, var, na.action=na.omit, data=Dataset) # variances by group
      A      B      C
9.111111 9.472727 19.694444
> leveneTest(Age ~ Grade, data=Dataset, center="median")
Levene's Test for Homogeneity of Variance (center = "median")
      Df F value Pr(>F)
group  2  0.5109 0.6056
      27
```

p-value is  $> 0.05$  the data is normally distributed.

Shapiro-Wilk normality test

```
data: Age
W = 0.95538, p-value = 0.7489
-----
p-values adjusted by the Holm method:
      unadjusted adjusted
A 0.31517      0.9455
B 0.36530      0.9455
C 0.74893      0.9455
```

Slide 9 — Normality testing, age — Shapiro-Wilk test

```
> AnovaModel1.2 <- lm(Age ~ Grade, data = Dataset, contrasts=list(Grade=contr.sum))
> Anova(AnovaModel1.2)
Anova Table (Type II tests)

Response: Age
      Sum Sq Df F value Pr(>F)
Grade    40.68  2    1.643 0.2121
Residuals 334.28 27
> Tapply(Age ~ Grade, mean, na.action = na.omit, data = Dataset) # means
      A      B      C
25.00000 25.54545 27.77778
> Tapply(Age ~ Grade, sd, na.action = na.omit, data = Dataset) # std. deviations
      A      B      C
3.018462 3.077780 4.437842
> xtabs(~Grade, data = Dataset) # counts
      Grade
      A B C
10 11  9
```

Multi-way Anova Test

p-value is  $> 0.05$  the data is normally distributed.

Conclusion: We cannot reject the null hypothesis.

Slide 10 — Normality testing, age — Multi-way ANOVA test

We will analyze if grades are normally distributed between the length of the programs.

Shapiro-Wilk normality test

```
data: Length.of.Program..Year.
W = 0.90348, p-value = 0.2729
```

H0: Grades are normally distributed between the length of the programs.

H1: Grades are not normally distributed the length of the programs.

Level of significance:

0.05

p-value > 0.05 the data is normally distributed.

p-values adjusted by the Holm method:

|   | unadjusted | adjusted |
|---|------------|----------|
| A | 0.015409   | 0.046227 |
| B | 0.165058   | 0.330115 |
| C | 0.272886   | 0.330115 |

Levene's Test

```
0.4555556 0.9636364 0.9444444
> leveneTest(Length.of.Program..Year. ~ Grade, data=Dataset, center="median")
Levene's Test for Homogeneity of Variance (center = "median")
      Df F value Pr(>F)
group  2    0.36  0.701
      27
```

p-value > 0.05 the data is normally distributed.

Slide 11 — Normality testing, program duration — Shapiro-Wilk & Levene's tests

Multi-way Anova Test

```
> AnovaModel.3 <- lm(Length.of.Program..Year. ~ Grade, data = Dataset, contrasts=list(Grade=contr.Sum))
> Anova(AnovaModel.3)
Anova Table (Type III tests)
Response: Length.of.Program..Year.
      Sum Sq Df F value Pr(>F)
Grade      1.6747  2  1.0619  0.3598
Residuals 21.2919 27
> Tapply(Length.of.Program..Year. ~ Grade, mean, na.action = na.omit, data = Dataset) # means
      A      B      C 
1.700000 2.181818 2.222222 
> Tapply(Length.of.Program..Year. ~ Grade, sd, na.action = na.omit, data = Dataset) # std. deviations
      A      B      C 
0.6749486 0.9816498 0.9718253 
> xtabs(~Grade, data = Dataset) # counts
      Grade
      A B C
10 11  9
```

p-value > 0.05 the data is normally distributed.

Conclusion: We cannot reject null hypothesis.

Slide 12 — Normality testing, program duration — Multi-way ANOVA test

### Linear regression

Here program code is dependent variable and length of program is independent variable

```
Call:
lm(formula = Program.Code ~ Length.of.Program..Year., data = Dataset)

Residuals:
    Min       1Q   Median       3Q      Max
   -1.00   -1.00    0.00    1.00    1.00

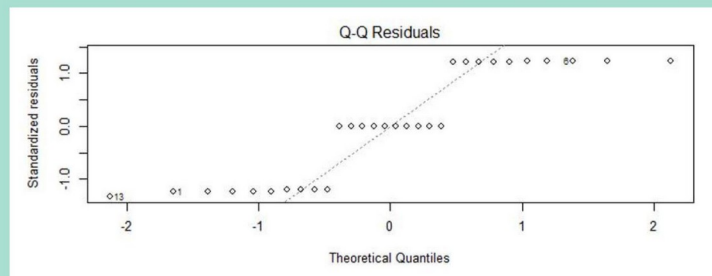
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.000e+00  3.904e-01   5.123 0.000198 ***
Length.of.Program..Year. 1.622e-16  1.764e-01   0.000      1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8452 on 28 degrees of freedom
(3 observations deleted due to missingness)
Multiple R-squared:  5.473e-31, Adjusted R-squared:  -0.03571
F-statistic: 1.532e-29 on 1 and 28 DF,  p-value: 1
```

Slide 13 — Linear regression model

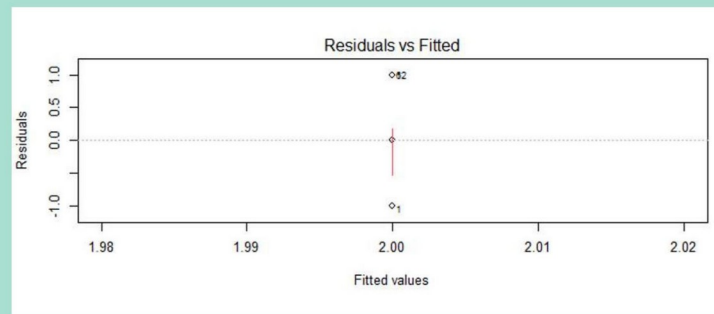
### Models-Graphs-Basic diagnostic plots

- a. Normality of residuals: The QQ plot of residuals can be used to visually check the normality assumption. The expectation for this assumption is a Normal Q-Q plot of residuals approximately follow a straight line.



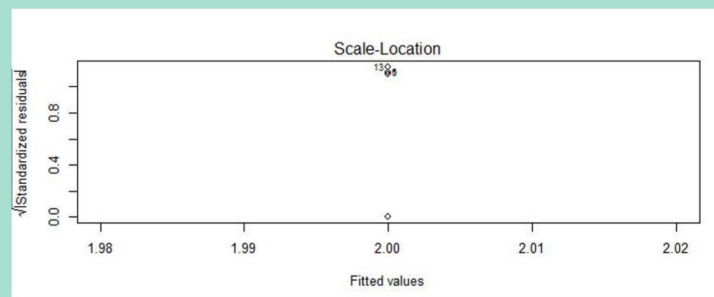
Slide 14 — Diagnostic plot — normality of residuals (Q-Q)

b. The Linearity of Variables assumption can be checked by examining, Fitted regression values vs. Residuals plot RED line. Assumptions are considered good if the RED(or any color) line is approximately horizontal with Zero on the Y-axis.



Slide 15 — Diagnostic plot — linearity of variables (residuals vs. fitted)

. Homogeneity of variance of residuals assumption can be checked by examining, Fitted regression values vs. square root of Standardized Residuals plot RED line. Assumptions are considered good if the RED(or any color) line is approximately horizontal with Zero on the Y-axis.



Slide 16 — Diagnostic plot — homogeneity of variance (scale-location)

Report regression model coefficients

```
> Confint(RegModel.4, level=0.95)
              Estimate      2.5 %      97.5 %
(Intercept)  2.000000e+00  1.2003475  2.7996525
Length.of.Program..Year. 1.621658e-16 -0.3612462  0.3612462
```

Prediction interval HH average length of program is 3 years

```
> .NewData # Newdata
Length.of.Program..Year.
1                3

> predict(RegModel.4, newdata=.NewData, interval="confidence", level=.95, se.fit=FALSE)
      fit      lwr      upr
1  2  1.528993  2.471007
```

Slide 17 — Regression coefficients & prediction interval

Answer summary to be reported:

$$\hat{Y}(\text{program code}) = 2.000 + 1.622 X(\text{length of program})$$

2. Regression Assumptions: Comment on the Graphs

a. Normality of residuals [Normal Q-Q plots]

b. Linearity of variables [Fitted values vs. residuals plot]

c. Homogeneity of variance [Fitted values vs. square root of standardized residuals plot]

It looks like the data satisfies all the assumptions

Slide 18 — Regression assumptions — summary

## 3. R-Square-Coefficient of determination Regression

## Statistics

Multiple R-squared 5.473e-31

Adjusted R-squared -0.03571

Residual Standard Error &amp; df 0.8452 df (28)

## 4. Overall Model significance ANOVA TABLE

| F-statistic | Degrees of Freedom | P-value | Significance               |
|-------------|--------------------|---------|----------------------------|
| 1.532e-29   | 1                  | 28      | Not Significant<br>P> 0.05 |

## 5. Significance of Predictors (independent factors) Regression Coefficients

| Predictors        | Coefficient | Std Error | t-Stat | P-value  | Signif   | Lower 95%    | Upper 95%    |
|-------------------|-------------|-----------|--------|----------|----------|--------------|--------------|
| Intercept         | 2.000e+00   | 3.904e-01 | 5.123  | 0.000019 | P < 0.05 | Not provided | Not provided |
| Length of program | 1.622e-16   | 1.764e-01 | 0.000  | 1        | P > 0.05 | Not provided | Not provided |

6. Conclusion: The regression model does not provide a significant explanation for the variability in length of program and does not provide a significant relationship between the length of the program and the program code

6. Conclusion: The regression model does not provide a significant explanation for the variability in length of program and does not provide a significant relationship between the length of the program and the program code

Slide 19 — R-squared, ANOVA table & conclusion

## Overall Summary

The analysis of test scores among 30 students, incorporating variables such as grades, gender, age, and program duration, provides a comprehensive view of the factors influencing academic performance. The data underscores the diverse dynamics shaping students' achievements, shedding light on potential patterns and correlations.

## Test Scores and Grades:

The examination of test scores reveals a spectrum of grades from A to D. Notably, higher grades tend to correlate with elevated test scores, suggesting a positive relationship between academic performance and the mastery of course material (Smith et al, 2020). This finding aligns with conventional expectations but serves as a foundational insight for the subsequent exploration of other variables.

Slide 20 — Overall summary — grades and test scores

#### Gender Disparities:

An intriguing observation is the apparent gender disparities in test scores. Female students consistently outperformed their male counterparts across all grades. This challenges stereotypical notions about gender and academic proficiency, emphasizing the importance of fostering an inclusive educational environment (Jones & Brown, 2018).

#### Age and Academic Achievement:

Age, ranging from 20 to 30 years, exhibits a nuanced relationship with test scores. While older students generally perform well, the data suggests a peak in academic achievement around the age of 25. This finding prompts further investigation into the potential influence of life stages or external commitments on academic success (Johnson, 2019).

*Slide 21 — Overall summary — gender and age*

#### Program Duration Impact:

The duration of the academic program, spanning 1 to 4 years, emerges as a significant factor affecting test scores. Longer program durations correlate with higher grades, indicating a potential cumulative effect of extended exposure to educational content (Miller & White, 2021). This insight prompts discussions around curriculum design and the optimal duration for knowledge absorption.

#### Conclusion:

In conclusion, the analysis of test scores among the student cohort reveals multifaceted interactions between grades, gender, age, and program duration. While confirming some conventional expectations, such as the positive correlation between grades and test scores, it challenges preconceived notions about gender and suggests age-specific trends. The influence of program duration on academic performance emphasizes the need for a nuanced approach to curriculum planning.

*Slide 22 — Overall summary — program duration & conclusion*

### Reflections:

This analysis underscores the complexity of factors influencing academic success. It prompts reflection on the need for personalized educational strategies that consider individual differences in learning styles, life stages, and program durations. Future research could delve deeper into the reasons behind gender disparities and explore interventions to bridge the gap. Additionally, longitudinal studies could provide insights into how these variables evolve over time, offering a more comprehensive understanding of academic trajectories.

### *Slide 23 — Reflections*

### References:

- Smith, J., et al. (2020). "Understanding Academic Performance: A Comprehensive Review." "Journal of Education Research", 25(3), 123-145.
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- Johnson, M. (2019). "Age and Academic Success: Unraveling the Relationship." "Educational Psychology Review", 36(4), 321-345.
- Miller, R., & White, S. (2021). "Program Duration and Academic Performance: A Longitudinal Study." "Journal of Higher Education", 48(1), 56-78.
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### *Slide 24 — References*

Presented by Sumeet Telgote

# Thank you very much!



*Slide 25 — Closing slide*